

Database Team 8 Post Office database Project

Our project was the Post Office database, where the goal was to design a website and database for a post office that could manage customers, employees, and packages. These requirements included a way to make an account, a way to create a package and track said package, and queries and reports detailing the data inside the database.

Our project was hosted on vercel for the frontend, render for the backend, and Azure Database for mysql flexible for the database. It was coded using nodejs for the backend, and Vite/React for the frontend. We used mysql workbench and mysql shell to work on the database on our personal computers, and github to collaborate on coding.

1a) Data that can be added

- New customer
- New employee
- New support ticket
- New package
- New package_pickup

1b) Data that can be edited

- Edit customer
- Edit employee
- Edit Support ticket
- Edit package

- Edit package_pickup

1c) Data that can be deleted

- Delete customer
- Delete employee

2) User roles

- Customer: Create support ticket, view personal packages, edit customer information
- Employee: View support tickets assigned to them, view and edit packages, add package, edit employee information, create/edit package_pickup
- Admin: View all support tickets, all employee privileges, create employee, view/delete employee, delete customer, Performance Report, Revenue Report, Ticket Report

3a) Triggers

- Trg_package_marked_lost: Lost package marked as lost and error message sent when a package is manually marked lost.
- Trg_shipment_marked_lost: All packages in a shipment marked lost if a shipment is marked lost.
- Trg_shipment_au_sync_pickup_arrival: When is shipment arrives at the Office, package_pickup is updated and fees are calculated

3b) Events

- Evt_mark_lost_packages: Packages that have not been updated for more than 14 days, and are not in a final status, are marked as lost.
- Evt_daily_package_pickup_storage: Calls stored procedure sp_daily_package_pickup_storage() at midnight to handle package pickup storage processing.

4a) Queries

- All customers: View customer information and their packages as well as delete customers
- Support Tickets: View support tickets information and edit ticket status/ resolution notes
- Packages: View all packages and edit status
- Employees: View employee information and delete employees

4b) Reports

- Revenue Report: View revenue by zone, package type, and quarterly breakdown based on the filter you apply.
- Performance Report: Compare two offices to find the best values of each, view best, worst, and office ratios of locations under the filters you apply.
- Ticket Report: View tickets by issue type and tickets by status according to the filters you apply.

5) Additional features

- Package tracking: find shipment history and delivery status of a package by its tracking number

- Package Price Calculator: Discover price of package by putting in package's information into the calculator.
- Package_pickup(employee/Admin only): Record a package being held at the office, and record a package being picked up from the office.